MTN Uganda - JUMO Integration

PA1 2019-08-27

Contents

Purpose 3

Scope 4

Assumptions 4

Solution Overview 5

Solution Integration and Use Cases 5

Registration 6

Prerequisites 6

Pre-condition 6

APIs 7

Request Loan 7

Prerequisites 8

Pre-condition 8

APIs 9

Request & Response Samples 9

Repay Loan 14

Prerequisites 15

Pre-condition 15

Request and Response Samples 15

Balance Check 17

Prerequisites 18

Pre-condition 18

Request and Response Samples 18

Loan repayment history 19

Prerequisites 19

Pre-condition 19

Request and Response Samples 20

GetMenuResourceInformation v1.0 21

Parameters 21

Session Traces 21

Security Description 26

Security Impact 27

Security Risk Management 27

Security Requirement 27

References 29

Security Approval 29

♥ Ericsson AB 2019
Commercial in confidence

Approval 29

Revision History

Revision	Date	Author	Description
PA1	25-07-2019	Ranjeet Singh Shekhawat	Initial version
PA2	25-09-2019	Ranjeet Singh Shekahwat	Updating as per the review comments.
PA3	23-10-2019	Ranjeet Singh Shekhawat	Updating as per the security review comments.
PA4	27-09-2020	Nehemiah Ogato	Updated flows
PA5	01-Oct-2020	Beegira Atuhaire	Add flows for sptransfer, status check and extra parameter clarification.

♥ Ericsson AB 2019
Commercial in confidence

1 Purpose

Purpose of this document is to provide a solution description for JUMO Saving and Loan use cases for USSD and Mobile App channels. The integration details for different systems involved in JUMO SNL use cases for USSD and Mobile App channels.

2 Scope

Scope of this document is to explain in more detail how requests from USSD and Mobile App (via Ericsson AM) will be handled and routed to different end systems for JUMO SNL use cases.

♥ Ericsson AB 2019
Commercial in confidence

3 Assumptions

- The existing mobile app server and EWP interface will be used to JUMO SNL use cases.
- MTN Uganda is on version 1.6 of Savings and Lending
- Customer journey will be served in English only, therefore language will default to “en”
- Any other existing interface will not be impacted.

4 Solution Overview

The Access Module node of EWP is integrated for front-end and back-end interface with JUMO server. The JUMO server is responsible for processing Saving and Loan requests for different banks.

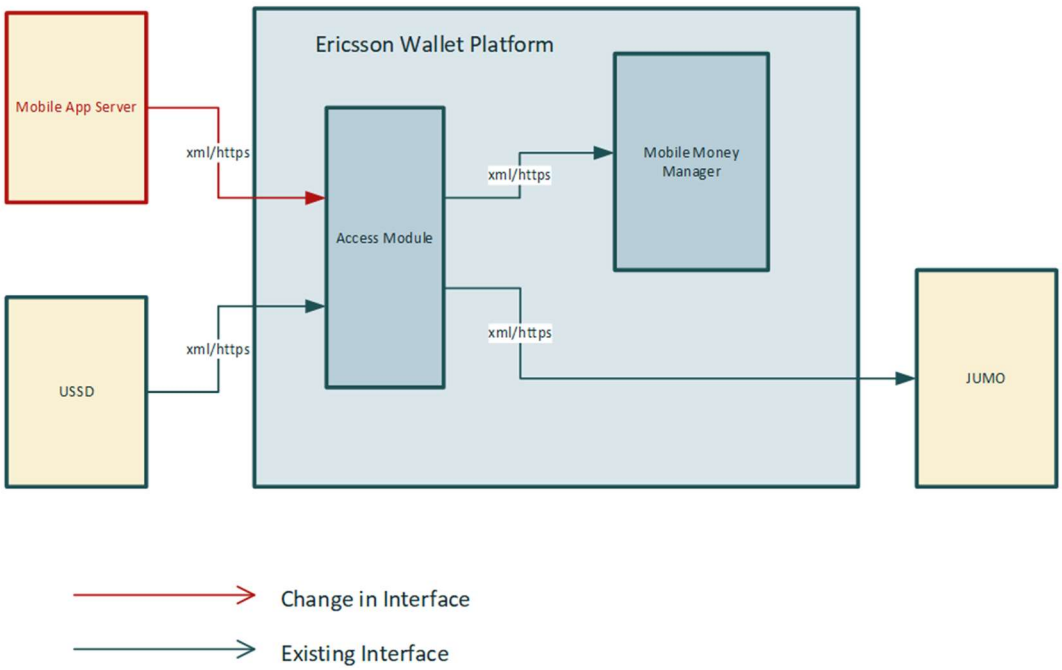


Figure 1 Architecture Diagram

The Mobile Money subscriber can access the MTN Mobile Money features using USSD and Mobile App. The JUMO SNL use cases are already accessible through USSD, however this document will ensure the same integration will be used for front-end via the new AM APIs.

4.1 Solution Integration and Use Cases

This section describes the interface between EWP, Mobile App and JUMO server. The access module is integrated with Mobile App server to handle the incoming xml over http APIs. The Access module also routes the incoming request to either Mobile Money Manager or JUMO Server using xml over http interface. The routing of incoming requests is based on use cases as described below.

♥ Ericsson AB 2019
Commercial in confidence

Complete USSD content to be configured is found in the attached document.
Variables are collected from JUMO as described in this document.

MTN Uganda - USSD - Loans - Acces

4.1.1 Registration

The subscriber needs to be registered on JUMO server to access the JUMO SNL sub-menu.
Registration flow will be synchronous with JUMO as JUMO currently uses in other markets.
The Journey:

I. Opt in (*Registration*) – Customer Journey

- Subscriber dials *165#
- Selects Savings & Loans
- Selects 'JUMO'
- A pop up with a link to the T&Cs is displayed and customer is prompted to enter Mobile Money pin to accept these and opt in for the service
- Upon a successful opt in, subscriber will receive;
    - An SMS from MTN confirming the successful opt in for example *'Y'ello you have successfully opted in for JUMO loans. Dial *165*5*2# to take a loan now.'*
    - Successful Opt in and account creation SMS notification from JUMO as well. o For cases of failed Opt in, customer should receive appropriate notification depending on the failure reason *(recycled MSISDNs, made the required time on MTN network, KYC)* for example *'Y'ello your request was unsuccessful. Visit a nearby MTN Service centre to update your KYC details.'*

4.1.1.1 Prerequisites

Access module should be configured with profile and login credentials in order to interact with the mobile money manager. JUMO registration is initiated by a user.

4.1.1.2 Pre-condition

The user should be in activated state, should have permission of getaccounts and add-preapproval.

♥ Ericsson AB 2019
Commercial in confidence

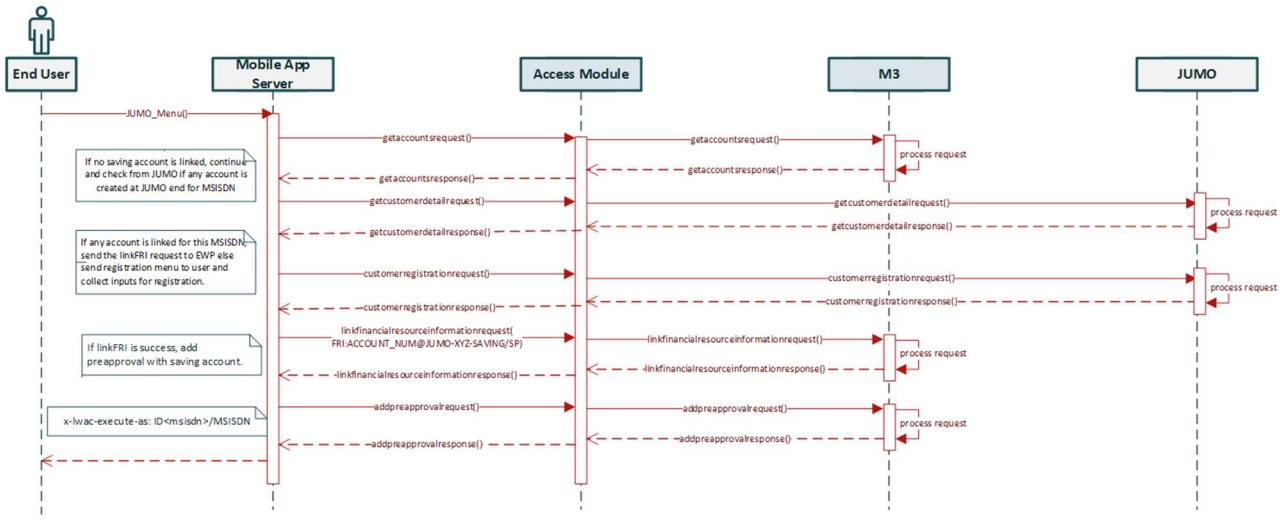


Figure 2 Sequence Flow - Registration

4.1.1.3 APIs

The below APIs are being received and sent to M3 and JUMO server.

Sl. No.	Client	Server	API	Authentication Type and Permission to Profile
1.	Mobile App Server	M3	Getaccounts1.0	System User authentication ROLE_GROUP_PERMISSIONS, ROLE_VIEW_ANY_ACCOUNT, ROLE_VIEW_ANY_ACCOUNT_DETAILS, ROLE_VIEW_CHILD_ACCOUNT, ROLE_VIEW_OWN_ACCOUNT
2.	Mobile App Server	JUMO	Getcustomerdetails	System User authentication
3.	Mobile App Server	JUMO	Customerregistration	System User authentication
4.	Mobile App Server	M3	Linkfinancialresourceinformation1.0	User authentication ROLE_UPDATE_OWN_ACCOUNTHOLDER
5.	Mobile App Server	M3	Addpreapproval	User authentication ROLE_UPDATE_OWN_ACCOUNTHOLDER

4.1.2 Request Loan

The subscribers who are registered on JUMO, can apply for Loan. The below sequence flow diagram explains the front-end's request and response added for both Mobile App and USSD and JUMO server.

♥ Ericsson AB 2019
Commercial in confidence

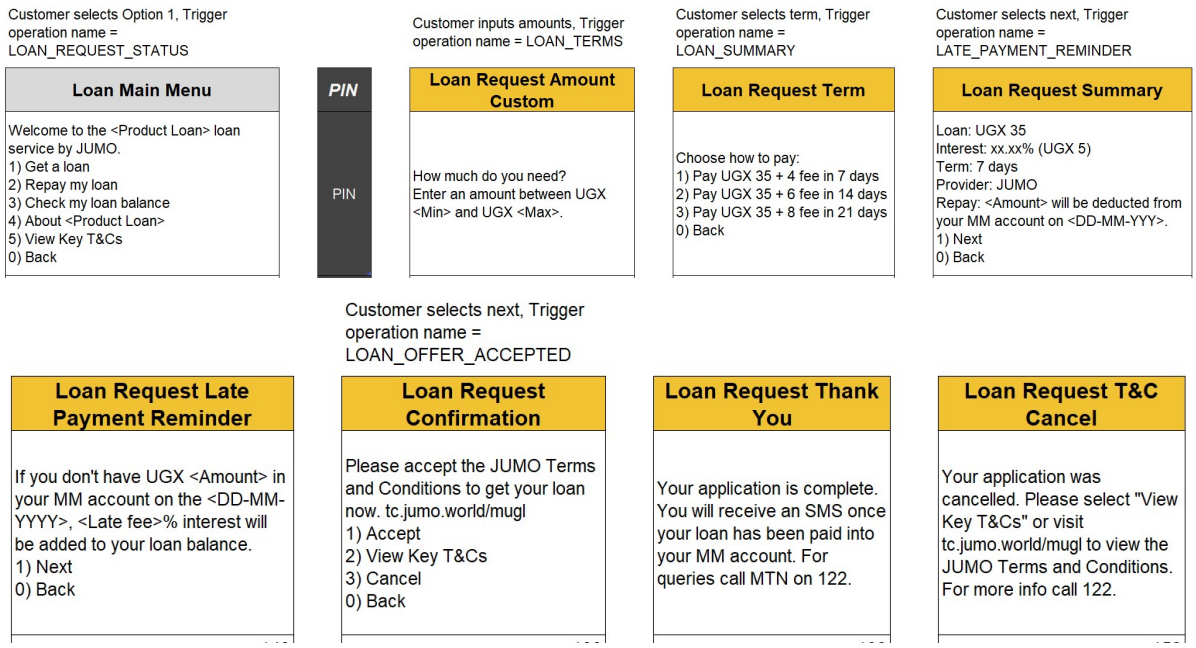
Also there is no change in backend interface for JUMO interface where JUMO Server triggers the request toward the ECW to complete the loan request.

4.1.2.1 Prerequisites

Access module should be configured with profile and login credentials in order to interact with the mobile money manager. Loan Request is initiated by a user.

4.1.2.2 Pre-condition

The user should be in activated state, should have permission of getaccounts.



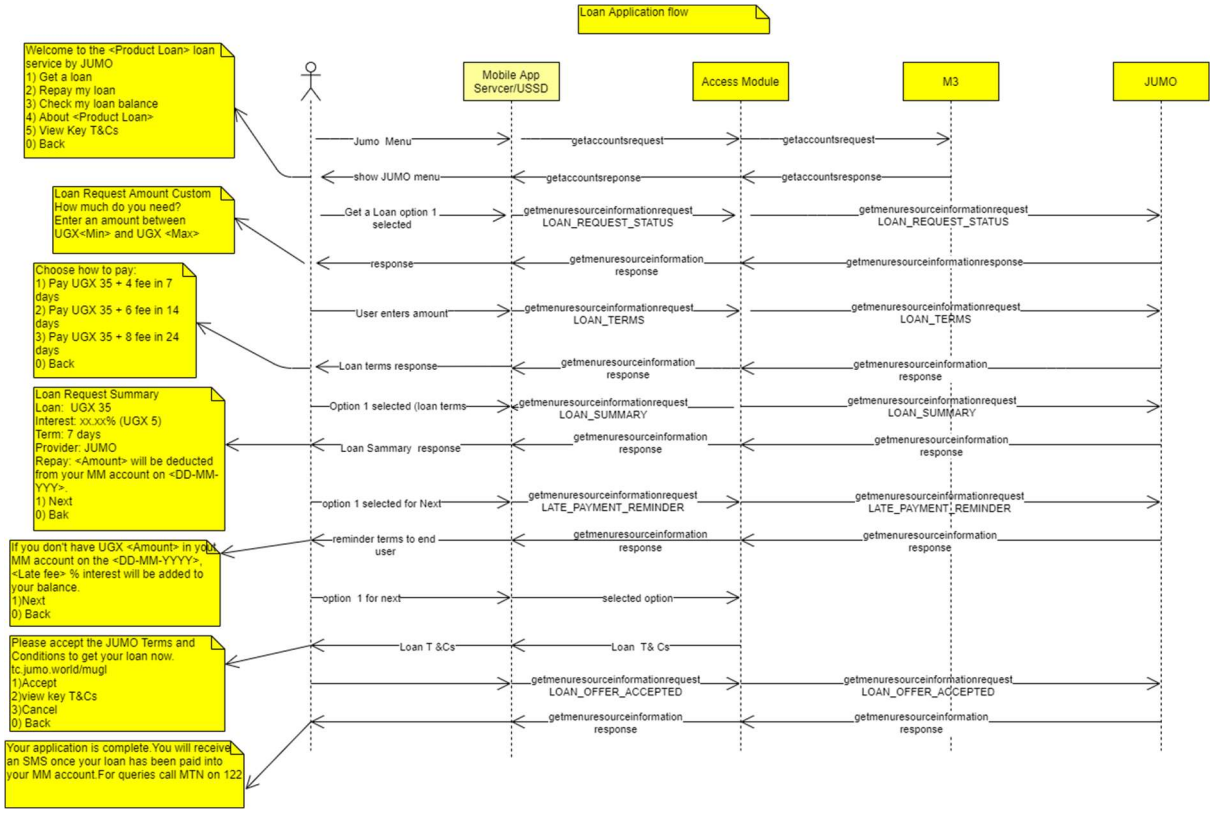


Figure 3 Sequence Flow - Loan Request

4.1.2.3 APIs

The below APIs are being received and sent to M3 and JUMO server from Mobile App server.

Sl. No.	Client	Server	API	Authentication Type
1.	Mobile App Server	M3	Getaccounts1.0	System User authentication ROLE_GROUP_PERMISSIONS, ROLE_VIEW_ANY_ACCOUNT, ROLE_VIEW_ANY_ACCOUNT_DETAILS, ROLE_VIEW_CHILD_ACCOUNT, ROLE_VIEW_OWN_ACCOUNT
2.	Mobile App Server	JUMO	Getmenuresourceinformation	System User authentication

4.1.2.4 Request & Response Samples

** Loan request status API **

API to be invoked immediately customer selects request loan sub menu.
operation name: LOAN_REQUEST_STATUS

Sample request:

```
< xml version="1.0" encoding="UTF-8" >
<p:getmenuresourceinformationrequest xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK">
  <resource>FRI:12345678@jumo.sp</resource>
    <sessionid>UUID</sessionid>
  <operationname>LOAN_REQUEST_STATUS</operationname>
  <extension>
    <language>en</language>
                    </extension>
  <channelinformation>USSD</channelinformation>
</p:getmenuresourceinformationrequest>
```

****Language will be sent as en (English) as the default for Uganda.**

Sample responses:

```
<p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK"
xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
    <message>SUCCESS</message>
    <extension>
      <minamount>10</minamount>
      <maxamount>1000</maxamount>
      <currency>GHS</currency>
  </extension>
</p:getmenuresourceinformationresponse>

<p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK"
xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
    <message>HAS_OTHER_OFFER</message>
    <extension>
      <minamount>10</minamount>
      <maxamount>1000</maxamount>
      <currency>GHS</currency>
  </extension>
</p:getmenuresourceinformationresponse>

<p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK"
xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
    <message>CUSTOMER_INELIGIBLE</message>
    <extension>
      <content>JUMO custom message to display</ content>
  </extension>
</p:getmenuresourceinformationresponse>

<p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK"
xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
    <message>HAS_EXISTING_LOAN</message>
    <extension>
      <outstandingbalance>1100</outstandingbalance>
      <currency>GHS</currency>
    </extension>
</p:getmenuresourceinformationresponse>
```

<message> possible values:

♥ Ericsson AB 2019
Commercial in confidence

- SUCCESS : Customer is eligible and qualifies for a loan offer for the product / bank selected. <extension> tag will contain currency, minimum loan and maximum loan limits they qualify for.
- HAS_OTHER_OFFER: Customer is eligible but DOES NOT qualify for a loan offer for the product selected. However the customer qualifies for an offer from another product / bank. The extension tag will have minimum, maximum loan limits they qualify for
- CUSTOMER_INELIGIBLE: Customer is not eligible for any loan offer. Content param used to display message to customer which is sent by JUMO
- LOAN_VELOCITY_LIMIT: Customer s loan request velocity violated hence cannot be offered loan at this time
- HAS_ANOTHER_LOAN_REQUEST: Customer has another loan request in processing state
- HAS_LOAN_REPAYMENT: Customer has another loan whose repayment is being processed
- HAS_EXISTING_LOAN: Customer has another active loan. <extension> tag will have total outstanding loan amount
- HAS_OVERDUE_LOAN: Customer has an overdue loan. <extension> tag will have total outstanding loan amount
- LOAN_REQUEST_DISABLED: JUMO has intentionally disabled loan request service

****JUMO to provide static responses for these menu items**

V. Loan Information

For a customer wishing to access additional product information;

- Customer dials *165#
- Selects 'Savings & loans'
- Selects 'JUMO'
- Selects 'Info'
- Customer will be able to access information on topics such as;
  - o About JUMO Loans
  - o Qualification
  - o Get bigger loans
  - o Fees
  - o Payment
  - o Late payment
  - o Key Terms & Conditions

** Loan terms API **

API to be invoked after the customer enters a validated loan amount. It expects the amount the customer entered and returns applicable loan terms for the loan amount applied for.
operation name: LOAN_TERMS

Sample request:

```
< xml version="1.0" encoding="UTF-8" >
<p:getmenuresourceinformationrequest xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK">
  <resource>FRI:12345678@jumo.sp</resource>      <sessionid>UUID</sessionid>
  <operationname>LOAN_TERMS</operationname>
  <extension>
    <amount>1000</amount>
 <extension/>
 <channelinformation>USSD</channelinformation>
</p:getmenuresourceinformationrequest>
```

Sample response:

```
<p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK" xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
```

```
    <message>SUCCESS</message>
    <extension>
      <loanamount>1000</loanamount>
      <currency>GHS</currency>
      <terms>
        <term>
        <termid>1</termid>
        <termdays>14</termdays>
        <termfees>50</termfees>
      </term>
      <term>
        <termid>2</termid>
        <termdays>21</termdays>
        <termfees>100</termfees>
      </term>
      </terms>
    </extension>
  </p:getmenuresourceinformationresponse>
```


** Loan summary API **

API to be invoked after the customer selects loan term. It expects the loan term id selected and returns details for preparing the loan request summary screen.

operation name: LOAN_SUMMARY

Sample request:


```
< xml version="1.0" encoding="UTF-8" >
<p:getmenuresourceinformationrequest xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK">
  <resource>FRI:12345678@jumo.sp</resource>
  <sessionid>UUID</sessionid>
  <operationname>LOAN_SUMMARY</operationname>
  <extension>
    <loanamount>1000<loanamount>
    <termid>2</termid>
      <providerid>1</providerid>
  <extension/>
  <channelinformation>USSD</channelinformation>
</p:getmenuresourceinformationrequest>
```


Sample response:


```
<p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK" xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
    <message>SUCCESS</message>
    <extension>
      <loanamount>1000</loanamount>
    <outstandingbalance>1100</outstandingbalance>
    <loanduedate>dd/mm/yyyy</loanduedate>
      <currency>GHS</currency>
    <feepercent>10</feepercent>
      <feeamount>100</feeamount>
      <provider>PROVIDER</provider>
  </extension>
</p:getmenuresourceinformationresponse>
```


** Late payment reminder API **

API to be invoked after the customer selects next on loan summary screen. It returns details for preparing the late payment reminder screen.

operationname: LATE_PAYMENT_REMINDER

Sample request:

```
< xml version="1.0" encoding="UTF-8" >
<p:getmenuresourceinformationrequest xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK">
  <resource>FRI:12345678@jumo.sp</resource>
   <sessionid>UUID</sessionid>
  <operationname>LATE_PAYMENT_REMINDER</operationname>
  <extension/>
  <channelinformation>USSD</channelinformation>
</p:getmenuresourceinformationrequest>
```

Sample response:

```
<p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK"
xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
   <message>SUCCESS</message>
   <extension>
      <loanamount>1000</loanamount>
     <totaloutstanding>1100</totaloutstanding>
     <loanduedate>dd/mm/yyyy</loanduedate>
      <currency>GHS</currency>
     <latepaymentfee>10</latepaymentfee>
      <tcsurl>https://protect-za.mimecast.com/s/xEDOC58MpWFZ96oZfylc1v</tcsurl>
  </extension>
</p:getmenuresourceinformationresponse>
```

** Loan offer accepted API **

API to be invoked after the customer confirms and accepts a loan offer.
operationname: LOAN_OFFER_ACCEPTED

Sample request:

```
< xml version="1.0" encoding="UTF-8" >
<p:getmenuresourceinformationrequest xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK">
  <resource>FRI:12345678@jumo.sp</resource>
   <sessionid>UUID</sessionid>
  <operationname>LOAN_OFFER_ACCEPTED</operationname>
  <extension/>
  <channelinformation>USSD</channelinformation>
</p:getmenuresourceinformationrequest>
```

Sample response:

```
<p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK"
xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
   <message>SUCCESS</message>
   <extension />
</p:getmenuresourceinformationresponse>
```

** Loan offer cancelled API **
API to be invoked after customer declines loan offer.
operationname: LOAN_OFFER_CANCELLED

Sample request:

```
< xml version="1.0" encoding="UTF-8" >
<p:getmenuresourceinformationrequest xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK">
 <resource>FRI:12345678@jumo.sp</resource>
   <sessionid>UUID</sessionid>
 <operationname>LOAN_OFFER_CANCELLED</operationname>
 <extension/>
 <channelinformation>USSD</channelinformation>
</p:getmenuresourceinformationrequest>
```


♥ Ericsson AB 2019
Commercial in confidence

Sample response:


```
<p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK"
xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
    <message>SUCCESS</message>
    <extension />
</p:getmenuresourceinformationresponse>
```


** Loan provider selection API **

API to be invoked when a customer qualifies for an offer from another loan provider . It expects loan terms selected and returns a list of providers / banks that a customer can select.

operation name: LOAN_PROVIDER_SELECTION

Sample request:


```
< xml version="1.0" encoding="UTF-8" >
<p:getmenuresourceinformationrequest xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK">
    <resource>FRI:12345678@jumo.sp</resource>
    <sessionid>UUID</sessionid>
   <operationname>LOAN_PROVIDER_SELECTION</operationname>
   <extension>
     <amount>1000<amount>
     <termid>2</termid>
     <language>en</language>
   <extension/>
   <channelinformation>USSD</channelinformation>
</p:getmenuresourceinformationrequest>
```


****Language will be sent as en (English) as the default for Uganda.**

Sample response:


```
<p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK"
xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
    <message>SUCCESS</message>
    <extension>
      <loanamount>1000</loanamount>
      <currency>GHS</currency>
      <providers>
        <provider>
          <id>1</id>
          <name>Provider One</name>
          <product>ProductOne</product>
        </provider>
        <provider>
          <id>2</id>
          <name>Provider Two</name>
          <product>ProductTwo</product>
        </provider>
      </providers>
    </extension>
</p:getmenuresourceinformationresponse>
```


4.1.3 Repay Loan

The subscribers who are registered on JUMO, can repay their loan. The below sequence flow diagram explains the front-end's request and response added for both Mobile App and USSD and JUMO server.

Also there is no change in backend interface for JUMO interface where JUMO Server triggers the request toward the ECW to complete the loan debit request.

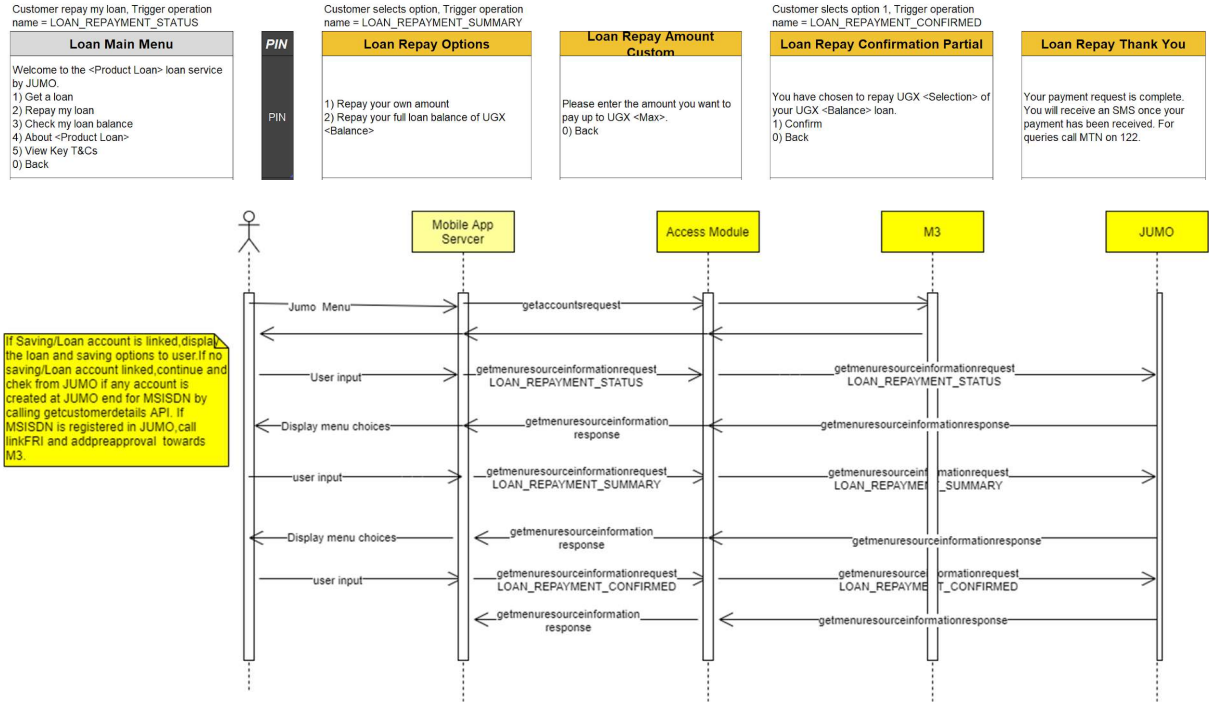
****Auto recoveries will follow the standard SNL Bankdebit flow.**

4.1.3.1 Prerequisites

Access module should be configured with profile and login credentials in order to interact with the mobile money manager. Repay Loan Request is initiated by a user.

4.1.3.2 Pre-condition

The user should be in activated state, should have permission of getaccounts.



4.1.3.3 Request and Response Samples

** Loan repayment status API **

API to be invoked immediately customer select repay loan menu.
operation name: LOAN_REPAYMENT_STATUS

Sample request:

```
< xml version="1.0" encoding="UTF-8" >
<p:getmenuresourceinformationrequest xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK">
  <resource>FRI:12345678@jumo.sp</resource>
 <sessionid>UUID</sessionid>
 <operationname>LOAN_REPAYMENT_STATUS</operationname>
 <extension>
  <language>en</language>
 </extension>
```

```
    <channelinformation>USSD</channelinformation>
  </p:getmenuresourceinformationrequest>
```

****Language will be sent as en (English) as the default for Uganda.**

Sample responses:

```
  <p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK"
xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
      <message>HAS_EXISTING_LOAN</message>
      <extension>
         <outstandingbalance>1100</outstandingbalance>
         <minamount>1</minamount>
         <currency>GHS</currency>
    </extension>
  </p:getmenuresourceinformationresponse>

  <p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK"
xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
      <message>NO_OUTSTANDING_LOAN</message>
      <extension />
  </p:getmenuresourceinformationresponse>

  <p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK"
xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
      <message>HAS_OTHER_OUTSTANDING_LOAN</message>
      <extension>
         <outstandingbalance>1100</outstandingbalance>
         <minamount>1</minamount>
         currency>GHS</currency>
         <provider>PROVIDER</provider>
    </extension>
  </p:getmenuresourceinformationresponse>
```

<message> possible values:

- HAS_EXISTING_LOAN : Customer has outstanding loan. <extension> tag will contain currency, outstanding balance and minimum loan repayment amount.
- NO_OUTSTANDING_LOAN: Customer has no outstanding loan
- LOAN_REPAYMENT_INPROGRESS: Customer as a loan repayment in progress
- HAS_OTHER_OUTSTANDING_LOAN: Customer has outstanding loan with another provider. <extension> tag will contain currency, outstanding balance, minimum loan repayment amount and provider/product name.

** Loan repayment summary API **

API to be invoked after the customer enters a validated repayment amount. The API expects the amount entered and returns the repayment amount requested and outstanding balance.

operation name: LOAN_REPAYMENT_SUMMARY

Sample request:

```
  < xml version="1.0" encoding="UTF-8" >
  <p:getmenuresourceinformationrequest xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK">
     <resource>FRI:12345678@jumo.sp</resource>
      <sessionid>UUID</sessionid>
     <operationname>LOAN_REPAYMENT_SUMMARY</operationname>
     <extension>
         <amount>500</amount>
      </extension>
     <channelinformation>USSD</channelinformation>
  </p:getmenuresourceinformationrequest>
```

Sample response:

♥ Ericsson AB 2019
Commercial in confidence

<p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK" xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
<message>SUCCESS</message>
<extension>
<repaymentamount>500</repaymentamount>
<outstandingbalance>1100</outstandingbalance>
<currency>GHS</currency>
</extension>
</p:getmenuresourceinformationresponse>

** Loan repayment confirmed API **

API to be invoked after customer confirms loan repayment.
operationname: LOAN_REPAYMENT_CONFIRMED

Sample request:

< xml version="1.0" encoding="UTF-8" >
<p:getmenuresourceinformationrequest xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK">
<resource>FRI:12345678@jumo.sp</resource>
<sessionid>UUID</sessionid>
<operationname>LOAN_REPAYMENT_CONFIRMED</operationname>
<extension/>
<channelinformation>USSD</channelinformation>
</p:getmenuresourceinformationrequest>

Sample response:

<p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK" xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
<message>SUCCESS</message>
<extension />
</p:getmenuresourceinformationresponse>

♥ Ericsson AB 2019
Commercial in confidence

4.1.4 Balance Check

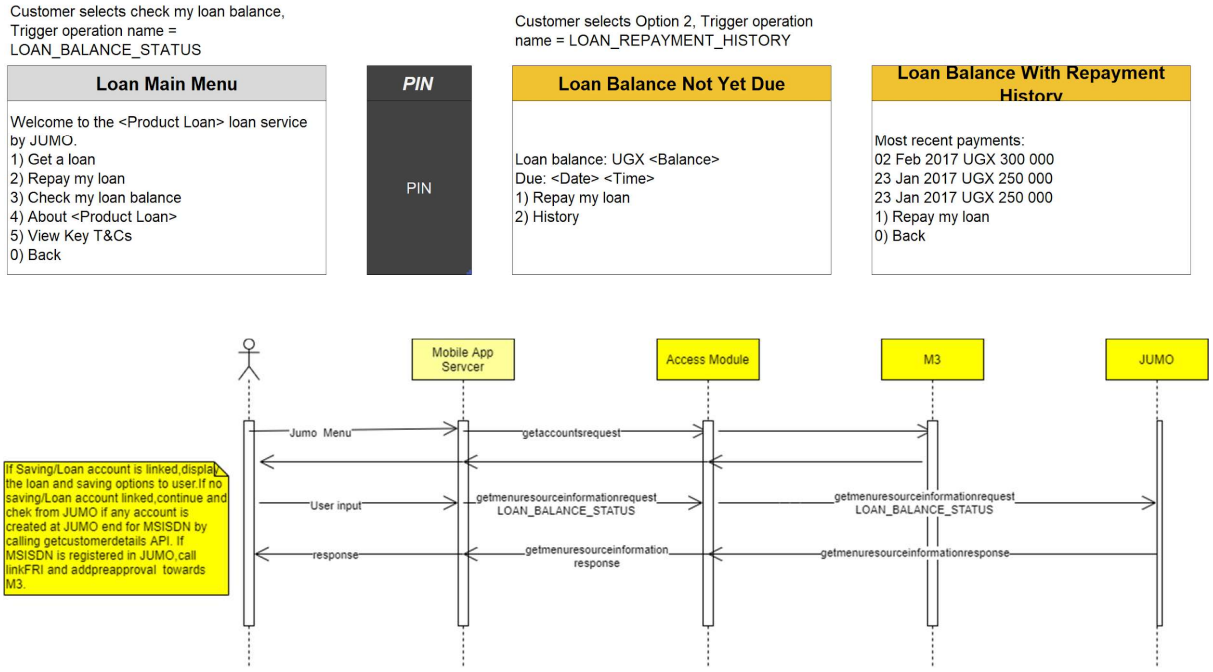
The subscribers who are registered on JUMO, can check their loan balances. The below sequence flow diagram explains the front-end's request and response added for both Mobile App and USSD and JUMO server.

4.1.4.1 Prerequisites

Access module should be configured with profile and login credentials in order to interact with the mobile money manager. Check loan balance Request is initiated by a user.

4.1.4.2 Pre-condition

The user should be in activated state, should have permission of getaccounts.



4.1.4.3 Request and Response Samples

** Loan balance status API **

API to be invoked after the customer selects check my loan balance menu.
operation name: LOAN_BALANCE_STATUS

Sample request:

```
< xml version="1.0" encoding="UTF-8" >
<p:getmenuresourceinformationrequest xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK">
  <resource>FRI:12345678@jumo.sp</resource>
    <sessionid>UUID</sessionid>
```

```
    <operationname>LOAN_BALANCE_STATUS</operationname>
    <extension>
      <language>en</language>
    </extension>
    <channelinformation>USSD</channelinformation>
  </p:getmenuresourceinformationrequest>
```


****Language will be sent as en (English) as the default for Uganda.**

Sample responses:


```
  <p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK" xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
      <message>SUCCESS</message>
      <extension>
         <outstandingbalance>1100</outstandingbalance>
         <duedate>2019/12/12 12:12:000</duedate>
         <currency>GHS</currency>
    </extension>
  </p:getmenuresourceinformationresponse>

  <p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK" xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
      <message>NO_OUTSTANDING_LOAN</message>
      <extension />
  </p:getmenuresourceinformationresponse>
```


<message> possible values:

- SUCCESS : Customer has outstanding loan. <extension> tag will contain currency, outstanding balance and loan due date.
- NO_OUTSTANDING_LOAN: Customer has no outstanding loan
- LOAN_REPAYMENT_INPROGRESS: Customer as a loan repayment in progress
- HAS_OTHER_OUTSTANDING_LOAN: Customer has outstanding loan with another provider. <extension> tag will contain currency, outstanding balance, loan due date and provider/product name

4.1.5 Loan repayment history

The subscribers who are registered on JUMO, can check their loan repayment history. The below sequence flow diagram explains the front-end's request and response added for both Mobile App and USSD and JUMO server.

API to be to be invoked when a customer selects a history option. Returns loan repayment history list if available.

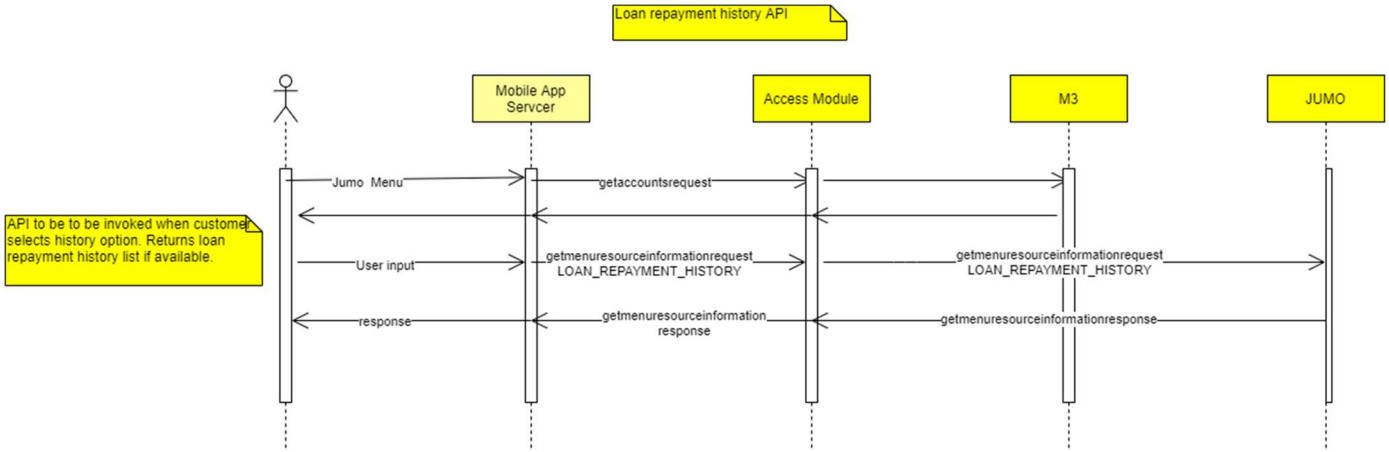
4.1.5.1 Prerequisites

Access module should be configured with profile and login credentials in order to interact with the mobile money manager. Check loan repayment history Request is initiated by a user.

4.1.5.2 Pre-condition

The user should be in activated state, should have permission of getaccounts.

♥ Ericsson AB 2019
Commercial in confidence



4.1.5.3 Request and Response Samples

** Loan repayment history API **

API to be invoked when a customer selects the history option. Returns loan repayment history list if available.
operation name: LOAN_REPAYMENT_HISTORY

Sample request:

```
< xml version="1.0" encoding="UTF-8" >
<p:getmenuresourceinformationrequest xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK">
  <resource>FRI:12345678@jumo.sp</resource>
  <sessionid>UUID</sessionid>
  <operationname>LOAN_REPAYMENT_HISTORY</operationname>
  <extension/>
  <channelinformation>USSD</channelinformation>
</p:getmenuresourceinformationrequest>
```

Sample responses:

```
<p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK"
xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
    <message>SUCCESS</message>
    <extension>
      <currency>GHS</currency>
      <repayments>
        <repayment>
          <amount>50</amount>
          <date>2019/12/12</date>
        </repayment>
        <repayment>
          <amount>75</amount>
          <date>2019/12/13</date>
        </repayment>
      </repayments>
  </extension>
</p:getmenuresourceinformationresponse>
```

```
<p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK"
xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
    <message>NO_REPAYMENT_HISTORY</message>
```

```
    <extension />
</p:getmenuresourceinformationresponse>
```

<message> possible values:

- SUCCESS : Customer has outstanding loan. <extension> tag will contain currency, outstanding balance and loan due date.
- NO_REPAYMENT_HISTORY: Customer has no loan repayment history

**Business to give exact descriptions and advise if the actual notifications to the end subscribers.

4.1.6 GetMenuResourceInformation v1.0

The getmenuresource API is being used for AM and JUMO interface.

Namespace: http://www.ericsson.com/em/am/ca/menu/v1_0/

Schema file: getmenuresourceinformation.xsd

4.1.6.1 Parameters

getmenuresourceinformationrequest

Name	Type	Optionality	Description
resource	string	Mandatory	This datatype respresents a Financial Resource Identifier of subscriber FRI:<MSISDN>@<spname>/SP. i.e. FRI:233558086960@XPRESS/SP
sessionid	string	Mandatory	Unique session identifier. i.e. 233558086960123456789
operationname	string	Mandatory	Operation name to identify type of request. i.e. LOAN_REQUEST_STATUS
channelinforation	communicationchannel	Mandatory	Front End channel, i.e. USSD, APP
locationinformation	locationinformation	Optional	Location information.
extension	extensiontype	Optional	Optional extension for sending extension data.

getmenuresourceinformationresponse

Name	Type	Optionality	Description
Message	string	Mandatory	Message contains the response type. Ie. SUCCESS or ERROR_RESPONSE
Extension	Extensiontype	Optional	Optional extension for sending extension data.

4.1.6.2 Session Traces

The below traces need to be captured by the Access Module once the user selects the JUMO submenu.

Name	Type	Optionality	Description
messageid	string	Mandatory	Unique request identifier i.e. MSISDN
sessionId	String	Mandatory	Unique session identifier

♥ Ericsson AB 2019
Commercial in confidence

sessionTrace	sessionTraceObject	Mandatory	A compound JSON object consisting: (Mandatory) ● userId - an identifier used to uniquely identify the user interacting with the front end channel e.g msisdn, ucid ● sessionId- used to uniquely identify distinct user sessions ● channel - front end channel the user interacted with. Can be USSD or MOBILE_APP ● startTime - session start time ● endTime - session end time ● transcript - A list consisting screen traversal details in chronological order. Each screen traversal includes: - screenName : Label / tag that uniquely identifies the screen within the journey - screenText: The actual text rendered to the user - userInput: The actual user input / selection while interacting with the screen - duration: Time spent by user on the screen (milliseconds) - finalScreen: A boolean flag denoting whether the screen was the last point of interaction with the user during the session

Sample Request


```
<?xml version="1.0" encoding="UTF-8"?>
<p:getmenuresourceinformationrequest xmlns:p="http://www.ericsson.com/em/am/ca/menu/v1_0/">
  <resource>FRI:233551423764@jumo.sp</resource>
  <sessionid>spwowlspowqlpl</sessionid>
  <operationname>SESSION_TRACE</operationname>
  <extension>
    <starttime>2019-03-15T13:29:49.503204</starttime>
    <endtime>2019-03-15T13:39:49.503204</endtime>
    <transcripts>
      <transcript>
        <duration>20</duration>
        <userinput>400</userinput>
        <screenname>LR</screenname>
        <screentext>How much are you requesting for? Enter amount between x and
          y.
        </screentext>
      </transcript>
      <transcript>
        <duration>6</duration>
        <userinput>1</userinput>
        <screenname>LC</screenname>
        <screentext>Confirm your repayment plan:\n1. xxxx in zz days\n0. Back</screentext>
      </transcript>
      <transcript>
        <duration>8</duration>
        <userinput>1</userinput>
        <screenname>LE</screenname>
        <screentext>Loan: 280\n Interest: x% (yyy) \nTerm: zz days Next\n0.
          Back\n
        </screentext>
      </transcript>
      <transcript>
```

```
        <duration>6</duration>
        <userinput>1</userinput>
        <screenname>LF</screenname>
        <screentext>If you don't have GHS xxx in your account on the 20 Apr 2019, a zz% late
          payment fee will be added to your loan balance.\n1.
          Next\n0. Back\n
        </screentext>
      </transcript>
      <transcript>
        <duration>15</duration>
        <userinput>1</userinput>
        <screenname>LG</screenname>
        <screentext>Please agree to get your loan now. \n1. Agree\n2. View terms\n0.
          Back\n
        </screentext>
      </transcript>
    </transcripts>
  </extension>
  <channelinformation>USSD</channelinformation>
</p:getmenuresourceinformationrequest>
```


Sample response:


```
  <p:getmenuresourceinformationresponse xmlns:p="https://protect-za.mimecast.com/s/fy_lC3l3n9upK2wpUg5ZmK"
xmlns:xsi="https://protect-za.mimecast.com/s/YTA4C482o9FB5lyBIxIM2d">
    <message>SUCCESS</message>
    <extension />
  </p:getmenuresourceinformationresponse>
```


4.1.7 Sptransfer

Used by the Service Provider to transfer money from the service provider's account to another registered (subscriber) account.

URI: /{baseContext}/sptransfer
Namespace: http://www.ericsson.com/em/emm/serviceprovider/v1_0/backend
Schema file: sptransfer.xsd

Parameters

sptransferrequest:			
Name	**Type**	**Optionality**	**Description**
sendingfri	string	Optional	The FRI of the sending account holder. If omitted, the default FRI of the sender is used. Validated with IsFRI.
receivingfri	string	Mandatory	The FRI of the receiving account holder. Parameter can not be NULL. Validated with IsFRI.

♥ Ericsson AB 2019
Commercial in confidence

amount	moneydetailstype v1.0	Mandatory	The amount to send from sender's account to receiver's account. Note: currently, the currency code must be the same as the currency code of the sender's account. Parameter can not be NULL. Validated with IsPositiveAmount.
providertransactionid	string	Mandatory	Service provider transaction id. Validated with NotBlank. Validated with Length (max=256). Validated with IsExternalReferenceString.
name	name v1.0	Optional	The name of the sender. This name will appear in the notification text sent to the receiver after money has been transferred to receiver's account.
sendernote	string	Optional	A descriptive note for sender's transaction history. Validated with Length (max=256). Validated with IsRestrictedString.
receivermessage	string	Optional	A descriptive message for receiver's transaction history. If present, it will also be added to the notification text sent to the receiver. Validated with Length (max=256). Validated with IsRestrictedString.
referenceid	string	Optional	A optional unique external reference identity. This identity must be unique for each request for the logged in service provider. If no response is sent back to the service provider due to a communication error etc, then the service provider can use this identity to ask about the status of the transfer. Validated with Length (max=64). Validated with IsExternalReferenceString.
quoteid	quoteid v1.0	Optional	The ID of the quote to use.

sptransferresponse:

Table 52 sptransferresponse

Name	Type	Optionality	Description

♥ Ericsson AB 2019
Commercial in confidence

transactionid	long	Mandatory	The financial transaction id of the transfer. Parameter can not be NULL.
loyalty	loyalty v1.0	Optional	The loyalty information for the sender.

Sample Request & response

Sample request

```
<?xml version="1.0" encoding="UTF-8"?>
<ns0:sptransferrequest
xmlns:ns0="http://www.ericsson.com/em/emm/serviceprovider/v1_0/backend">
  <sendingfri>FRI:JUMO.sp/USER</sendingfri>
  <receivingfri>FRI:256775666852/MSISDN</receivingfri>
  <amount>
    <amount>600000.00</amount>
    <currency>UGX</currency>
  </amount>
  <providertransactionid>6003215</providertransactionid>
  <sendernote>Loan Refund</sendernote>
  <receivermessage>Redemption of voucher</receivermessage>
  <referenceid>4577132456</referenceid>
</ns0:sptransferrequest>
```

Response

```
<?xml version="1.0" encoding="UTF-8"?>
<ns6:sptransferresponse
xmlns:ns6="http://www.ericsson.com/em/emm/serviceprovider/v1_0/backend">
  <transactionid>3051661</transactionid>
</ns6:sptransferresponse>
```

4.1.8 Gettransactionstatus (optional)

This API is used to query the status of get status of previously initiated Payment, Transfer or Debit transaction initiated by a 3PP. This API may be triggered in case a response for a completed request isn't received from the mobile money platform.

URI: /{baseContext}/gettransactionstatus
Namespace: http://www.ericsson.com/em/emm/financial/v1_0
Schema file: gettransactionstatus.xsd

Parameters

gettransactionstatusrequest:

Table 494 gettransactionstatusrequest

Name	Type	Optionality	Description
referenceid	string	Mandatory	Reference ID which has been entered by user, while initiating the transaction. Parameter can not be NULL. Validated with Length (min=1) (max=64). Validated with IsExternalReferenceString.
identity	string	Optional	A valid identity associated with the account holder that performed the requested transaction. Validated with IsIdentity.

gettransactionstatusresponse:

Table 495 gettransactionstatusresponse

Name	Type	Optionality	Description
transactionid	long	Mandatory	The financial transaction ID. Parameter can not be NULL.
status	financialtransactionstatus v1.0	Mandatory	Payment Status. Parameter can not be NULL.

Sample Request:

```
<ns2:gettransactionstatusrequest xmlns:ns2="http://www.ericsson.com/em/emm">
  <referenceid>1192053-1-0-256779999771-500</referenceid>
</ns2:gettransactionstatusrequest>
```

Sample response:

```
<ns2:gettransactionstatusresponse xmlns:ns2="http://www.ericsson.com/em/emm">
  <financialtransactionid>1192053-1-0-256779999771-500</financialtransactionid>
            <status>SUCCESSFUL</status>
</ns2:gettransactionstatusresponse>
```

Get Transaction Status Error Response:

```
<ns2:errorResponse xmlns:ns2="http://www.ericsson.com/lwac"
errorcode="TRANSACTION_NOT_FOUND"/>
```


♥ Ericsson AB 2019
Commercial in confidence

5 Security Description

5.1 Security Impact

TBD.

5.2 Security Risk Management

S/N	Risk Description	Rating	Vulnerability	Affected Component	Mitigation
	N/A		•		

5.3 Security Requirement

The security guidelines for this demand aligns with E// security standard. All existing security requirements applies.

Security Area	Domain	Description	Current Configuration/Evidence
Client Authentication	Access Control	Either of the following should be supported at minimum for client authentication. - normal authentication (API session based). Authentication of clients is performed by a username/password-based API login service - via the basic authentication scheme as defined in RFC 1945	The integration is for outgoing requests. Client Authentication = normal authentication.
Mutual Authentication support	Access Control	TLS based Mutual authentication must be supported i.e. Both Client and server must authenticate each other by presenting trusted certificates before allowing any network connection	TLS 1.2 enabled integration.
tIntegrity Protection for Data in transit	Data protection	Software shall support message integrity through digital signature verification and creation. The signature algorithm such as RSA encryption over an SHA-256 digest of the signed data must be supported.	TLS 1.2 enabled integration.
Strong encryption methods	Data Protection	If encryption is used for securing the data (rest, or in motion), the key	

♥ Ericsson AB 2019
Commercial in confidence

		lengths and algorithms shall be "strong" by current definitions. Any weak or broken encryption methods should not be used such as DES, MD5, etc	
Confidentiality & Integrity of data in motion	Network Data Protection	All communication interfaces internal or external must support secure protocols such as TLS (TLS 1.2 or above) to protect confidentiality and integrity of the messages.	TLS 1.2 is supported.
Certificates issued by a CA	Network Data Protection	System must support use of X509 certificates for TLS based connectivity issued by an external certificate authority (CA preference must be agreed with operator). External CA must be trusted by system.	M3-external-ca signed certificate for TLS setup.
Static IP addresses	Network Data Protection	The application should support use of static IP's for communication. If there is a strong need for dynamic IP's for communication, domain name resolution must be done only against an authorized domain names/systems with support for DNSSEC.	Static IP interface.
List of Api.		All API's existing or new should be stated and listed	Customize API: Getcustomerdetail Customerregistration getmenuresourceinformation Note: System user authentication will be used for the customize APIs. The User credential will be provided by Jumo. Xml over http APIs Getaccounts1.0 Linkfinancialresourceinformation1.0 Addpreapproval 1.0
List of profiles and permissions		All new profiles and permissions should be stated and listed	The below are existing profiles - System User Profile: "MTNGH Lwam" User Profile: "MTNGH Enhanced KYC Subscriber Profile" "MTNGH Medium KYC Subscriber Profile" "MTNGH Minimum KYC Subscriber Profile" Permissions:

			profiles.zip

6 References

[A] JUMO Customer Journey - https://erilink.ericsson.se/eridoc/erl/objectId/09004cffc28248a3?docno=&action=current&format=excel12book

7 Security Approval

Ericsson:

Name	Role	Signature	Comment	Date
	Solution Architect			
Simon Agaba	Security Architect			
	Customer Project Manager			

♥ Ericsson AB 2019
Commercial in confidence

8 Approval

This document has been mutually reviewed and approved to constitute the full scope of the solution to be delivered by the MTN Mobile Money Project. This document is signed in two (2) copies of which the parties have taken one (1) each.

MTN

Abdul Ssebagala

Date (YYMMDD): 2020.10.03

Place: Kampala

Signature: .....

Ericsson

Beegira Atuhaire/Nehemiah Ogato

Date (YYMMDD): .....

Place: .....

Signature: .....

Beegira Atuhaire

Digitally signed by Beegira Atuhaire
Date: 2020.10.05 09:13:04 +03'00'

JUMO

Andrea Oxenham

Date (YYMMDD): 03/10/2020

Place: Cape Town, South Africa

Signature: